14 · Case 9, part 2: label the synapse, and why levodopa

The words of this track, as it says them. Bold brown is the examiner.

Exam case 9, continued. Label the synapse, and why levodopa.

In her session she said you will get this specific drawing in the exam. And in another: they're going to ask you to pinpoint the presynaptic neuron, and label it. So these are free marks, if the letters are learned.

Nine marks. Label her synapse figure, A to G, and name the two transport processes it shows.

The hook: vesicles mean pre. Receptors mean post.

Letter A is a calcium channel, where calcium enters the cell.

B is a vesicle, which contains the neurotransmitters.

C is the neurotransmitter molecules. Every dot is one.

D is a reuptake channel, where the neurotransmitter goes back, to be used again.

E is the receptors, on the postsynaptic membrane.

F is the presynaptic neuron. You can tell by the vesicles.

G is the postsynaptic neuron. You can tell by the receptors.

Then the two processes. Exocytosis: the vesicle fuses with the membrane, and releases the neurotransmitter. And diffusion: the neurotransmitters cross the synaptic cleft, from high concentration to low.

Her formative test words the same figure as drop-downs. B is a vesicle containing neurotransmitters, such as serotonin, epinephrine or dopamine. E is a receptor on the postsynaptic membrane. The process moving neurotransmitters into the cleft is exocytosis. And the process moving them across the cleft is simple diffusion.

Three marks. Levodopa travels through the blood and into the brain, and is used as medication. Why not dopamine itself?

To work on brain cells, a medication must be able to cross the blood-brain barrier. Substances that aren't lipid soluble, fat soluble, can't cross it, so they're useless for treating conditions of the brain. And the third line, which is the textbook's rather than hers: levodopa crosses into the brain, and is made into dopamine there. Dopamine itself doesn't cross the blood-brain barrier.

Her Kahoot makes the same point with alcohol: alcohol affects the brain because it can cross the blood-brain barrier.

Once more. The synapse labels, and levodopa.

Letter A, a calcium channel. B, a vesicle of neurotransmitter. C, the neurotransmitter molecules. D, a reuptake channel. E, the receptors. F, the presynaptic neuron, with the vesicles. G, the postsynaptic neuron, with the receptors. Into the cleft by exocytosis, and across it by diffusion. Levodopa, not dopamine, because a drug must cross the blood-brain barrier to work on the brain. Levodopa crosses and is made into dopamine there; dopamine itself can't cross.